

Day 38 - clustering in Machine Learning

1) K-Means clustering - Mathematical Implementation

K-Means clustering is an unsupervised machine learning algorithm used to divide data into K clusters.

It groups similar data points together.

Main Idea

- choose the number of clusters (K)

- Find the center of each cluster (centroid)

- Assign each data point to the nearest centroid

- Update centroids again and again until clusters become stable

Steps of K-Means

- choose number of clusters K

- select random centroids

- calculate distance of each point from centroid

- Assign each point to nearest centroid

- Recalculate centroid

- Repeat until no major change happens

Distance Formula (Euclidean Distance)

To find the nearest centroid:

$$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$$

This formula measures the distance between two points.

Mathematical Objective of K-Means

K-Means tries to minimize the Within cluster sum of squares (WCSS).

Formula:

$$J = \sum_{i=1}^k \sum_{x \in c_i} \|x - \mu_i\|^2$$

Where:

k = number of clusters

c_i = cluster i

x = data point

μ_i = centroid of cluster i

Meaning:

The algorithm tries to make points inside a cluster as close as possible to their centroid.

Simple Example

Suppose we have points:

$(1,2), (2,3), (8,9), (9,10)$

If $K=2$, then K-Means may form:

cluster 1:

$(1,2), (2,3)$

cluster 2:

$(8,9), (9,10)$

Now centroids become:

centroid 1

$$((1+2)/2, (2+3)/2) = (1.5, 2.5)$$

centroid 2

$$((8+9)/2, (9+10)/2) = (8.5, 9.5)$$

This process repeats until clusters become fixed.

Advantages of K-Means

- Easy to understand

- Fast and simple

- Works well for spherical clusters

Disadvantages

Need to choose K manually

Sensitive to outliers

Not good for irregular-shaped clusters

2) How Hierarchical clustering Works

Hierarchical clustering is an unsupervised learning algorithm used to create a hierarchy or tree of clusters.

It forms clusters step by step.

Types of Hierarchical clustering

1. Agglomerative (Bottom-Up)

Starts with each point as a separate cluster

Merges the closest clusters step by step

2. Divisive (Top-Down)

Starts with all data points in one cluster

Splits them into smaller clusters

Most commonly used:

Agglomerative Hierarchical clustering

How Agglomerative Hierarchical clustering Works

Steps:

consider each point as a separate cluster

Find the two nearest clusters

Merge them

Repeat until all points are merged into one big cluster

Dendrogram

A dendrogram is a tree-like diagram used to show how clusters are merged.

Use of Dendrogram:

Helps decide the number of clusters

Shows similarity between data points

Example

Suppose points are:

A, B, C, D

Initially:

ABCD

Then if A and B are closest:

$(A, B)CD$

Then if C and D are close:

$(A, B)(C, D)$

Finally:

$((A,B),(C,D))$

This creates a hierarchical tree.

Advantages

- No need to specify K at the start

- Easy to visualize using dendrogram

- Good for small datasets

Disadvantages

- Slow for large datasets

- Hard to undo wrong merges

3) DBSCAN - Implementation and Importance

DBSCAN stands for:

Density-Based Spatial clustering of Applications with Noise

It is an unsupervised clustering algorithm that groups data points based on density.

Main Idea of DBSCAN

Points that are close together and have many nearby points form a cluster.

Points that do not belong to any dense region are treated as noise (outliers).

Important Parameters in DBSCAN

1. Epsilon (ϵ)

The maximum distance between two points to be considered neighbors.

2. MinPts

Minimum number of nearby points required to form a dense region.

Types of Points in DBSCAN

1. Core Point

A point having at least MinPts neighbors within ϵ .

2. Border Point

A point close to a core point but not having enough neighbors itself.

3. Noise Point

A point that does not belong to any cluster.

How DBSCAN Works

Choose ϵ and MinPts

Find neighbors of each point

Identify core points

Expand clusters from core points

Mark isolated points as noise

Simple Example

Suppose a group of points are very close together → DBSCAN makes them one cluster.

If one point is far away from everyone, it is marked as:

Noise / outlier

Importance of DBSCAN

DBSCAN is important because:

- It can find clusters of any shape

- It can detect outliers

- It does not require specifying number of clusters in advance

- Works better than K-Means for irregular cluster shapes

Advantages

- Detects outliers automatically

- Works on irregular-shaped clusters

- No need to define K

Disadvantages

Choosing ϵ and MinPts can be difficult

Not ideal when cluster densities vary a lot

4) What is GMM? How it Works

GMM stands for Gaussian Mixture Model.

It is an unsupervised probabilistic clustering algorithm.

Unlike K-Means, GMM does not assign a point to only one cluster directly.

Instead, it gives the probability of belonging to each cluster.

Main Idea of GMM

GMM assumes that the data is made up of multiple Gaussian (normal) distributions.

Each cluster is represented by one Gaussian distribution.

How GMM Works

Suppose there are 2 clusters.

For a data point, GMM may say:

80% chance it belongs to cluster 1

20% chance it belongs to cluster 2

So GMM gives soft clustering, while K-Means

gives hard clustering.

Difference Between K-Means and GMM

K-Means

Each point belongs to only one cluster.

GMM

Each point has a probability of belonging to multiple clusters.

How GMM Works Step by Step

GMM uses the Expectation-Maximization (EM) Algorithm.

Step 1: Initialization

Start with random Gaussian parameters.

Step 2: Expectation (E-Step)

calculate probability that each point belongs to each cluster.

Step 3: Maximization (M-Step)

Update cluster parameters:

- Mean

- Variance

- Weight

Step 4: Repeat

Repeat E-Step and M-Step until model becomes

stable.

Advantages of GMM

- Gives probability-based clustering

- Better than K-Means for overlapping clusters

- Works well when clusters are elliptical

Disadvantages

- More complex than K-Means

- Slower

- Needs proper parameter tuning

5) What are the Evaluation Metrics Used for clustering Algorithms?

Evaluation metrics are used to check how well the clustering algorithm has grouped the data.

Since clustering is unsupervised, evaluation is slightly different from classification.

Common Evaluation Metrics for clustering

1. Silhouette Score

Silhouette Score measures how similar a data point is to its own cluster compared to other

clusters.

Range:

-1 to +1

Meaning:

+1 → very good clustering

0 → overlapping clusters

-1 → poor clustering

Interpretation:

Higher silhouette score means better clustering.

2. Inertia / WCSS

WCSS means Within cluster sum of Squares.

It measures how close the points are to their cluster centroid.

Goal:

Lower WCSS means better clustering.

Used in:

K-Means clustering

3. Davies-Bouldin Index (DBI)

This metric measures the average similarity between clusters.

Interpretation:

Lower value = better clustering

Higher value = poor clustering

4. Calinski-Harabasz Index

This metric measures:

separation between clusters

compactness inside clusters

Interpretation:

Higher value = better clustering

5. Adjusted Rand Index (ARI)

Used when true labels are available.

It compares actual labels with predicted clusters.

Range:

-1 to 1

Interpretation:

Higher ARI = better clustering

6. Normalized Mutual Information (NMI)

Also used when actual labels are known.

It measures how much information is shared

between:

true classes

predicted clusters

Interpretation:

Higher NMI = better clustering